a

Tiantan Hospital Cohort for Model Development (N = 8,670, 2017-2022)

Static Clinical Data	Age, Sex, Diagnosis, Surgery
Vital Signs	HR, MAP, Temperature
Laboratory Tests	Na^+, K^+, Glucose, Creatinine
Medication	Drug Type, Dose, Timing
Fluid Balance	Input, Output

Internal Test (N = 2,073, 2023)

Patients$_i$

Data from Tiantan Hospital

External Test (N = 819)

Patients$_i$

Data from Wuhan Union Hospital

b

MIMIC-IV
Medication-lab events

Drug$_i$ Drug$_j$ Drug$_d$

Dose events Lab windows

$$\Delta Y_{p,l}^{(\tau)} = Y_{p,l}(t+\tau) - Y_{p,l}(t^-)$$

$w_{i,1}^{(k,\tau)}$ $w_{i,2}^{(k,\tau)}$ $w_{i,m}^{(k,\tau)}$ $w_{j,1}^{(k,\tau)}$ $w_{j,2}^{(k,\tau)}$ $w_{j,m}^{(k,\tau)}$

Dose level Time windows

25% K%

$$e_{d\to l} = \left[w_{d,l}^{(k,\tau)}\right]_{k\in K,\tau\in T}$$

Patient Lab Test Indicators Drug Intervention

MIMIC-IV Effect Graph #Stage 1

Na Desmopressin K Thiazide Dextrose 5% Cl Urea

Effect Spectrum:
one intervention → many labs

Unit / time-step conversion
prior matrices B & A

Neurosurgical Subgraph #Stage 2

Glucose (Next day) Glucose Na Na (Next day) Desmopressin

Temporal Propagation:
Intervention→next-day change→labs return

knowledge prior vectors

GNN

Construction and Transfer of Medication Effect Maps Based on MIMIC-IV Real-World Data

c

24-hour rolling landmark prediction

Patient observations

Update at t = 24, 48, 72, ...

Landmark *t*

No future information used

6-h actionable warning gap

24-h prediction window

48-h prediction window

(t+6, t+30)

(t+30, t+54)

t *t+6* *t+30* *t+54*

d

Patient observations

Dual-scale Temporal Backbone

Local branch

Selective SSM

Local State

Multi-resolution interaction

Clinical Event Tokenizer

Event tokens

Causal Transformer

Global State

Global branch

Drug-effect prior vectors

Patient-conditioned Graph-Prior Adapter

Prior Embedding

K, V

Q

Spare cross-attention

Adaptive knowledge gate

Graph-prior State

Fusion

e

Multi-Task Output layer

Tiantan next 24h (AUROC) Wuhan Union next 48h (AUROC) Tiantan next 48h (AUROC) Wuhan Union next 24h (AUROC)

AUROC 0.829 0.823 0.793 0.802

AUPRC 0.496 0.540 0.337 0.298

Dynamic hyponatremia risk prediction at 24- and 48-hour horizons

Paired patient-cluster bootstrap: top three models

Macro-F1

p=0.116 ns p=0.451 ns p=0.503 ns

NaBSIGHT LR-L2 LightGBM CNN1D RandomForest CatBoost KNN MLP-static TCN XGBoost GRU-D SVM-linear Transformer BalanceRF LSTM AdaBoost BiGRU-attn ExtraTrees GRU LR-ElasticNet HistGB RETAIN MLP-sklearn BiLSTM BiGRU

Ordinal hyponatremia severity prediction at 24- and 48-hour horizons

Tiantan Hospital | temporal test | next 24 h — MAE = 1.92, RMSE = 2.56, R^2 = 0.68

Tiantan Hospital | temporal test | next 48 h — MAE = 2.14, RMSE = 2.85, R^2 = 0.53

Wuhan Union Hospital | external | next 24 h — MAE = 2.26, RMSE = 2.98, R^2 = 0.67

Wuhan Union Hospital | external | next 48 h — MAE = 2.48, RMSE = 3.33, R^2 = 0.54

Predicted minimum sodium (mmol/L) Observed minimum sodium (mmol/L)

Future serum sodium nadir prediction at 24- and 48-hour horizons